THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 1 km/h is equal to: 1

(a) 18/5 m/s **(b)** 3.6 m/s **(c)** 1 m/s **(d)** 5/18 m/s

2. An ancient device that uses flowing water to measure time is a: 1

(a) sand watch **(b)** quartz clock **(c)** water clock **(d)** sundial

3. A bus moving at 40 km/h takes how much time to cover 200 km? 1

(a) 0.2 h **(b)** 8 h **(c)** 5 h **(d)** 4 h

4. 36 km/h in m/s is: 1

(a) 3.6 m/s **(b)** 36 m/s **(c)** 100 m/s **(d)** 10 m/s

5. The oscillations of which crystal are used in modern watches? 1

(a) diamond **(b)** ruby **(c)** salt **(d)** quartz

6. An object covers equal distances in equal intervals of time. Its motion is: 1

(a) non-uniform **(b)** random only **(c)** uniform **(d)** periodic only

7. 20 m/s in km/h is: 1

(a) 20 km/h **(b)** 72 km/h **(c)** 36 km/h **(d)** 7.2 km/h

8. Which of the following is a periodic motion? 1

(a) falling of a stone **(b)** a running car

(c) flying of a bird **(d)** swinging of a pendulum

9. An object covers unequal distances in equal intervals of time. Its motion is: 1

(a) at rest **(b)** non-uniform **(c)** circular only **(d)** uniform

10. Which of these is a unit of speed? 1

(a) m/s **(b)** kg **(c)** m **(d)** s

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a pendulum used in clocks? 2

12. A car travels 150 km in 3 h. Find its speed. 2

13. What do odometer and speedometer measure? 2

14. Find the time taken to cover 90 km at 45 km/h. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the differences between uniform and non-uniform motion with examples. 3

16. What is the principle of a quartz clock? 3

17. Name three situations where measuring time precisely is essential. 3

OR

Draw (describe) the distance-time graph of an object moving with uniform speed. 3

18. What is the time period of the second hand of a clock? 3

OR

Find the average speed of a car that travels 60 km in the first 2 h and 40 km in the next hour. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The distance-time table is: 0 s 0 m; 2 s 10 m; 4 s 20 m; 6 s 30 m. Describe the motion and find the speed. 5

OR

A bus travelling 45 km/h takes 4 h for a journey. Find the distance and the time needed for the same distance at 60 km/h. 5

20. Explain how an odometer and a speedometer work in a vehicle, and use readings to find average speed. 5

OR

Explain periodic and non-periodic motion with examples in nature. 5

END OF QUESTION PAPER — Total: 40 Marks

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